



U8: Evolution

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abrupt appearance	the sudden emergence of a new species or group of animals in the fossil record
adapt	to adjust or change to suit conditions
adaptation	inherited characteristic/trait that increases an organism's chance of survival
adaptive radiation	the rapid, divergent evolution of multiple species from a single common ancestor, driven by adaptation
analogous structures	structures that have the same function but different construction; not evidence of a common ancestor
anatomical homology	similar body structures among different species from a common ancestor
ancestor	an early species of animal or plant from which other species have evolved
artificial selection	selective breeding of plants and animals to promote the occurrence of desirable traits in offspring
biogeography	study of the distribution of organisms around the world
bottleneck effect	a reduction in the genetic diversity of a population caused by a decrease in size due to environmental events
cladogram	diagram with branches that represent the hypothesized phylogeny or evolution of a species or group
coevolution	process in which two or more species evolve in response to changes in each other
common ancestor	the most recent species from which two different species evolved
competition	the struggle between organisms to survive in a habitat with limited resources



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convergent evolution	type of evolution where unrelated species independently evolve similar traits but live in different parts of the world; driven by similar environmental pressures
descended from (a descendant)	comes from a common ancestor
developmental homology	similarities in the development of embryos between different species that share a common ancestor
directional selection	type of natural selection that increases the expression of the extreme versions of a trait in a population
disruptive selection	type of natural selection that removes individuals with average traits but retains individuals expressing extreme traits at both ends of the continuum
divergent evolution	evolution of one or more closely related species into different species; resulting from adaptations to different environmental conditions
embryo	a multicellular organism in the earliest stage of development; pre-birth
embryology	study of embryos and their development
environment	the surroundings or conditions in which a person, animal, or plant lives or operates.
environmental resource	materials provided by the earth for use; water and sunlight are examples
evolution	cumulative changes in a groups of organisms over time
evolutionary theory	explains the diversity of life where species change over time, share common ancestors, and adapt to their environment
fitness	the ability of an organism to survive and reproduce in its environment



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fossil record	chronological collection of life's remains in rock layers of the earth
founder effect	change in allele frequencies as a result of the migration of a small subgroup of a population
gene flow	movement of alleles from one population to another
gene / allele frequency	how often a particular gene or allele occurs in the population
gene pool	all the genes, including all the different alleles for each gene, that are present in a population at any one time
genetic drift	a change in the allele frequency of a population due to random chance events
genetic variation	diversity in DNA among individuals in a population
genome map	a record of the entire genome of an organism
gradualism	theory that evolution occurs slowly and steadily over time
habitat	physical area in which an organism lives
homologous structures	structures that are anatomically similar in different species; evidence of common ancestry (the same down to their bones)
inherited variation	genetic differences passed down from parents to offspring
isolation	separation from others
molecular homology	similar amino acid sequences (DNA) among different species; evidence of common ancestry
native	an organism that occurs naturally in a particular region, ecosystem, or habitat (without human intervention)
natural selection	



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	the survival and reproduction of individuals due to their inherited traits, leading to changes in the population (aka evolution)
natural resistance	an organism's inherent ability to withstand threats like pathogens or antibiotics
non-random mating	occurs when individuals in a population choose mates based on specific traits
phylogenetics	the study of evolutionary relationships among species or population based on shared ancestry
population	group of individuals of the same species that live in the same area
predator	an animal that hunts other animals for food
prey	an animal that is hunted and killed by another for food
punctuated equilibrium	the theory that species evolve during short periods of rapid change followed by long periods of stability
recombination	a combining of genes or alleles different from what they were in the parents; takes place during meiosis
reproductive isolation	separation of a species or population so that they can no longer interbreed and eventually evolve into two separate species
reproductive success	the passing of genes to the next generation who also pass those genes to their offspring
speciation	the formation of new and distinct species
species	a group of organisms that can interbreed and produce fertile offspring
stabilizing selection	type of natural selection that favors average individuals over extreme variations; reduces genetic diversity



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stasis

a long period of little to no evolutionary change

transitional fossils

fossils that show links in traits between groups of organisms; used to document intermediate stages in the evolution of a species

vestigial structure

a structure that is present in an organism but no longer serves its original purpose